Predicting Toronto Home Sale Prices: Two Valuation Scenarios

SSU Data Dash 2026

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Two valuation scenarios

Off-market

No asking price. The level has to come from the home and from sales that have already closed.

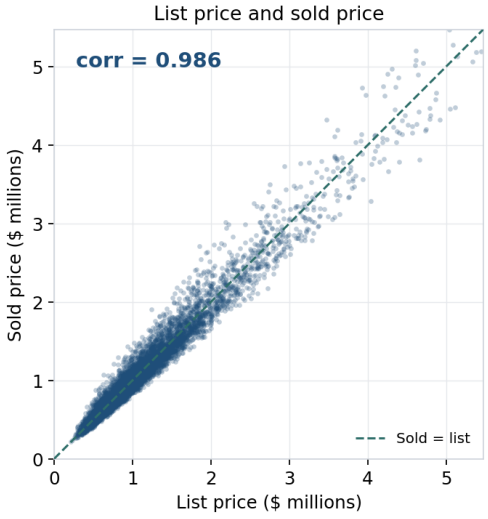
Target $\log Y$

List-aware

The asking price L is known. The target is the gap around that ask.

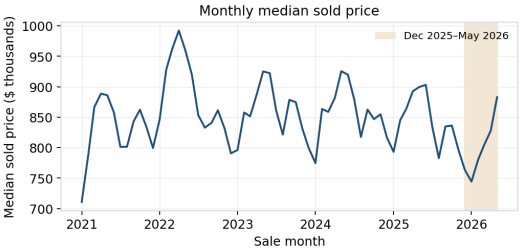
Target $\log(Y/L)$

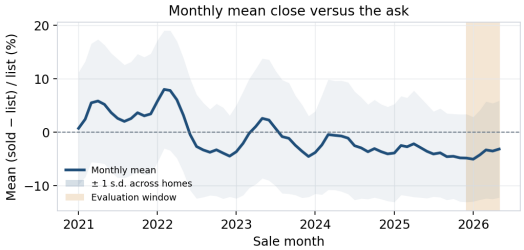
Y is the sold price. The dollar price is the ask times the predicted ratio.



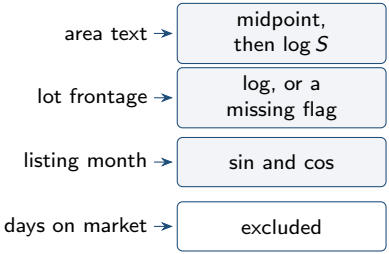
Dataset overview and market dynamics

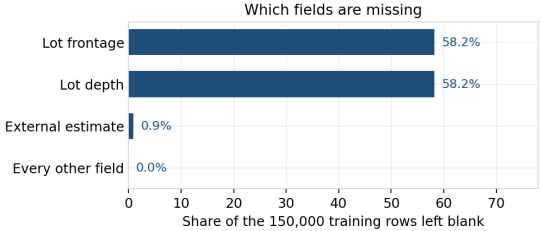
150,000 synthetic Toronto sales, January 2021–May 2026. Attributes, location, listing text, dates, and prices. No external data. Full-book median sold/list ratio 0.98.





Data preparation and feature engineering





The lot blanks are the same rows. They are the condominiums.

Off-market model development

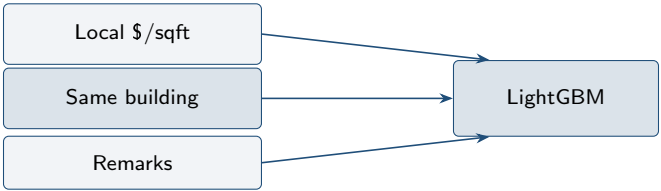
$$A_i = S_i \, m_{c(i),t(i)}(\tau_i)$$

m: median sold \$/sqft, same community and type, 12 months before listing. Fewer than 30 sales falls back to the type, then the city.

Formulation	MdAPE
Historical anchor	15.60%
LightGBM on the anchor residual	7.54%
Direct log-price LightGBM	7.53%

One fit through November 2025, scored on December 2025–May 2026. Not the walk-forward refit.

Off-market: historical sales and property information



Illustrative comparable. Not a fitted home.

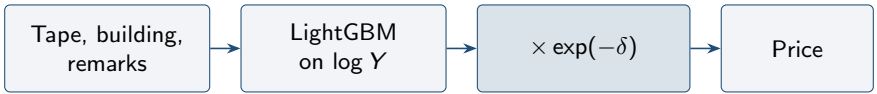
Earlier $/sqft	Index then	Index now	Brought forward	Column
$640	100	108	$691	$\log(S \times 691)$

Index $= \exp(I_{t,m})$, I a three-month median of earlier log $/sqft. Coverage 59.8%. Remarks are 0.7% of split gain.

Off-market architecture and calibration

$$\hat{Y} = \exp\big(F(x)\big)\ \exp(-\delta), \qquad \delta = \text{median}_{j \in \text{recent}}\big(\log \tilde{Y}_j - \log Y_j\big)$$

A positive δ is recent overprediction. The price is scaled by $\exp(-\delta)$.



July–November 2025 screen	MdAPE	Median signed
No correction	8.31%	+4.71%
Full correction	**7.97%**	+3.77%

The three months are before the origin, and the tree that measures δ did not see them. This screen chose the weight. It is not the walk-forward claim.

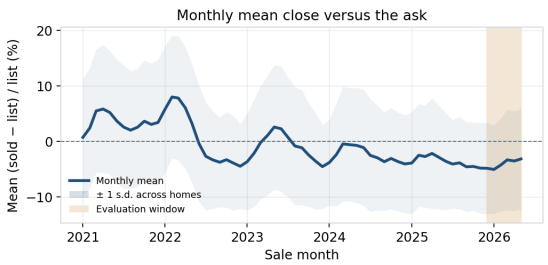
LightGBM is the engine on both tracks: it splits on categories, missing values, and interactions among the home and the comps. On an earlier list-aware feature set it matched XGBoost and beat a random forest.

List-aware: the gap around the asking price

The ask carries the level. A log-linear fit of $\log Y$ on $\log L$ and the attributes has elasticity 0.894 and $R^2 = 0.966$.

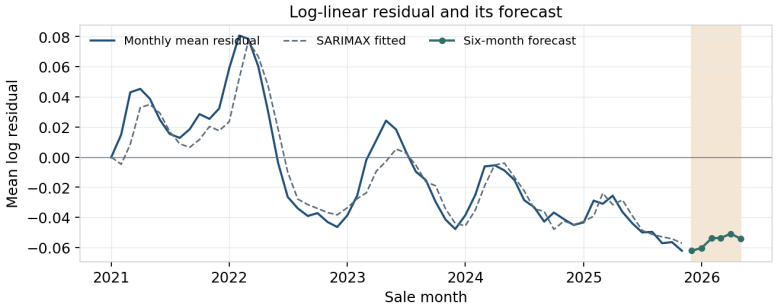
$$g_i = \log(Y_i/L_i), \qquad \hat{Y}_i = L_i \exp(\hat{g}_i).$$

Full-book median $Y/L = 0.98$, about 2% under the ask. The monthly mean of that gap is not flat.



Modeling temporal residuals with SARIMAX

$$\log Y_i = \beta_0 + \beta_L \log L_i + z_i^\top \gamma + e_i \qquad (1 - \phi L)(1 - \Phi L^{12})\bar{e}_m = c + u_m$$

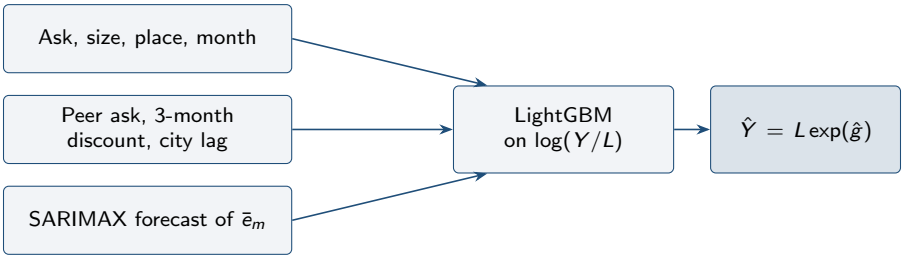


Fitted order $\mathrm{SARIMAX}(1,0,0) \times (1,0,0,12)$. The product of the two autoregressions includes a lag-13 term.

Putting that forecast on as a hard monthly shift, on the log-linear model alone, moves developmental MdAPE from 7.13% to 4.78% and the median signed error from $+5.95\%$ to $+0.25\%$. The tree does not apply that shift after the fact.

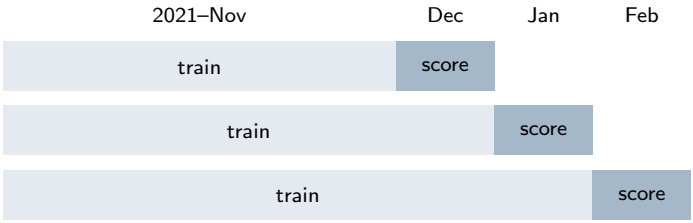
List-aware model specification

$$\hat{Y} = L \exp(\hat{g})$$



The forecast is an input. The tree can use less of it on a home the monthly factor does not fit. A further shift after the tree had walk-forward signed error -1.75% and was left off.

Walk-forward validation

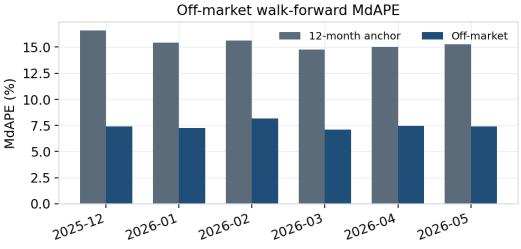


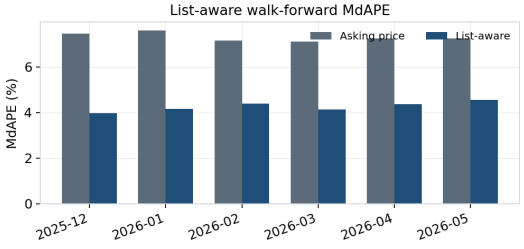
March, April, and May repeat the step: the training bar grows by the month just scored. July–November 2025 was used only to choose the specification.

10,069 sales. Tables, the SARIMAX, the ridge, the tree, and δ are rebuilt on closes before the month being scored. A random split would put later closes into the comps.

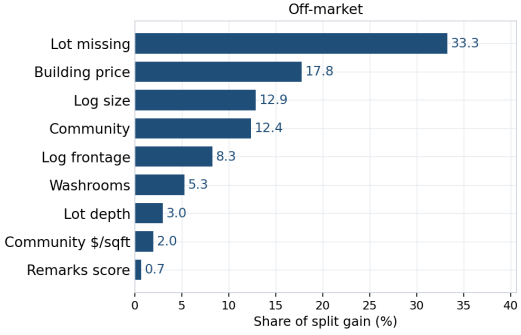
Walk-forward performance

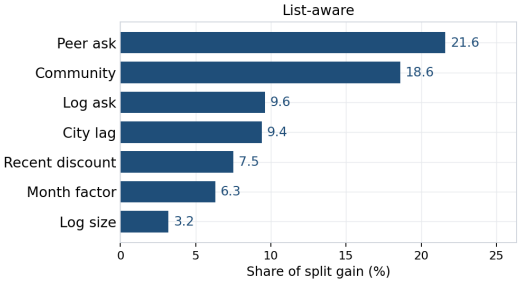
	Baseline MdAPE	Model MdAPE	MAPE	Median signed
Off-market	15.33%	**7.45%**	9.79%	−0.37%
List-aware	7.26%	**4.28%**	5.37%	+0.90%



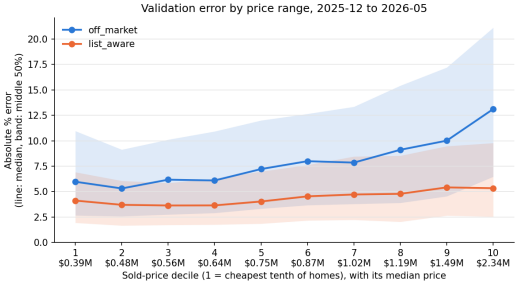


Feature importance





Prediction errors by price



Absolute error: median, and the middle 50%.

Stale tape, May: MdAPE 7.39% to 8.38%, signed $+0.98\%$ to -5.29%.



Signed error. Same 10,069 sales and the same sold-price cuts. Positive means the prediction is high.

Conclusions and limitations

Off-market

7.45% MdAPE

15.33% baseline, a 51% reduction

Direct log-price prediction with historical comparable sales and recent-error calibration.

List-aware

4.28% MdAPE

7.26% baseline, a 41% reduction

Log closing ratio, with local market indicators and a SARIMAX feature.

Remaining limitations

Coverage

40% lack an earlier same-building sale. 46% lack a recent local discount.

Systematic bias

High-end underprediction off-market. Persistent positive bias, list-aware.

Uncertainty

Synthetic data, point estimates, and dependence on recent historical sales.

Historical market information improves both valuation tasks, but accuracy depends on the availability and recency of comparable evidence.